

The $M/G/\infty$ Queue

Natarajan Gautam
Department of Industrial and Systems Engineering
Texas A&M University
235A Zachry, College Station, TX 77843-3131
Email: gautam@tamu.edu
Phone: 979-845-5458
Fax: 979-847-9005

July 9, 2009

Abstract

The objective of this chapter is to describe the $M/G/\infty$ system where the arrivals are according to a Poisson process, the service times are independent and identically distributed, and there are an infinite number of servers. We analyze the number in the system under both transient and steady state conditions. We describe results for the busy period distribution of as well as departure from an $M/G/\infty$ queue. We finally study an extension in terms of considering time-varying arrivals.

An $M/G/\infty$ queue is one where the arrivals are according to a Poisson process and the service time for each arriving entity is generally distributed with cumulative distribution function (CDF) $G(\cdot)$. Further, there are an infinite number of servers in the system and therefore the sojourn time in the system equals the service time. Let λ be the average arrival rate. Also, let τ denote the mean service time. We will typically consider the mean service time to be finite, i.e. $\tau < \infty$. Also, for non-triviality we assume $\tau > 0$. It is crucial to note that there is no restriction in terms of the service time random variable, it could be discrete, continuous or a mixture of discrete and continuous.

The first time anyone hears about the $M/G/\infty$ queue, the biggest concern is the issue of infinite number of servers. It does not seem practical and if it is a purely theoretical exercise, why bother? Several reasons come to mind. There are systems where infinite servers are excellent approximations such as conveyor belts where for all practical purposes the entities on the conveyor belt can begin service as soon as they arrive. Similarly systems with *self-service* can also be modeled as $M/G/\infty$ queues. A more common approximation is when the number of servers in an $M/G/c$ queue is large and the traffic intensity is not too high (i.e. not almost 1). Typically the $M/G/c$ queue for $c > 1$ is difficult to analyze. Hence the $M/G/\infty$ queue is also known as a queue with Poisson arrivals and *ample servers*.

Other motivations for studying $M/G/\infty$ queues are to obtain insights into the model and analysis that can be used in related systems. For example to explain some of the results of

$M/G/c/c$ queues, one typically turns to $M/G/\infty$ results. Moreover, among all the systems where the arrival or service (or both) are non-exponential, $M/G/\infty$ is the only one that can be analyzed for transient behavior, departures from queues and time-varying parameters. These reasons are usually sufficient to make a case to teach $M/G/\infty$ in courses and also include in textbooks (and encyclopedias).

Recently there have been some renewed interest and new results for $M/G/\infty$ while being applied to modeling traffic in computer-communication systems. Parulekar and Makowski [8] consider a slotted-time system where the arrivals in a given slot were proportional to the number in the system of an $M/G/\infty$ queue with heavy tail service time (such as a Pareto distribution). Notice that since the number in the system for an $M/G/\infty$ is correlated over time, using a Pareto distribution results in long-range dependence which is shown to be satisfied in traffic in computer-communication systems. Chang [1] characterizes such an arrival process using large-deviations theory to obtain the tail distribution of subsequent buffers.

Further, Likhanov and Mazumdar [7] as well as Resnick and Samorodnitsky [9] extend the slotted (or discrete time) version developed by Parulekar and Makowski [8] to the continuous time case. In particular, requests arrive according to a Poisson process and every request is responded by a heavy-tailed response where fluid is poured at a constant rate. These are excellent models for web servers where information flows out (response) to requests that arrive according to a Poisson process and file sizes are heavy tailed. These are referred to as fluid- $M/G/\infty$ queues. In summary, the $M/G/\infty$ queue can be used to suitably approximate a number of applications and there is a need to understand them in order to obtain useful insights.

Having motivated the need to study $M/G/\infty$ queues, here is how the remainder of this chapter is organized. In Section 1 we describe transient and steady state analysis of the number in the system. We use those results to derive the busy period distribution in Section 2. Then in Section 3, we study the departures from $M/G/\infty$ queues. We consider an extension to the standard $M/G/\infty$ system by allowing time-varying arrivals in Section 4.

1 Number in system: Transient and Steady-State

Consider an $M/G/\infty$ queue having independent and identically distributed service times with CDF $G(\cdot)$. Let $\{N(t), t \geq 0\}$ be a Poisson process with parameter λ that counts the number of arrivals in time $(0, t]$ for all $t \geq 0$. Define $X(t)$ as the number (of entities) in the system at time t . The objective of transient and steady-state analysis is to obtain a probability distribution of $X(t)$ for finite t and as $t \rightarrow \infty$ respectively. We would first present the transient analysis and then take the limit for steady-state analysis.

Transient analysis typically depends on the initial state of the queue. To obtain simple closed-form expressions, we need to make one of the following three assumptions: (i) the queue is empty at $t = 0$, i.e. $X(0) = 0$; (ii) the queue started empty in the distant past, i.e. $X(-\infty) = 0$, and hence the stochastic process $\{X(t), t \geq 0\}$ is stationary; (iii) if $X(0) > 0$, then the service for all the $X(0)$ entities *begin* at $t = 0$. If one of the above three assumptions are not satisfied, we will have to know the times when service began for each of the $X(0)$

customers in order to obtain a distribution for $X(t)$.

For the transient analysis here, we make the first assumption, i.e. $X(0) = 0$. Hence we are interesting in computing $p_j(t)$ defined as

$$p_j(t) = P\{X(t) = j | X(0) = 0\}.$$

Note that if we made the second assumption, then $X(t)$ would be according to the steady state distribution for all $t \geq 0$. However if we made the third assumption, then

$$P\{X(t) = j | X(0) = i, B_i\} = \sum_{k=0}^{\min(i,j)} \binom{i}{k} [1 - G(t)]^k [G(t)]^{i-k} p_{j-k}(t),$$

with the event B_i denoting that service for all i initial customers begins at $t = 0$. Since this is straightforward once $p_j(t)$ is known, we continue with obtaining $p_j(t)$ by making the first assumption.

Notice that $\{X(t), t \geq 0\}$ is a regenerative process with regeneration epochs corresponding to when the queueing system becomes empty. To obtain $p_j(t)$, consider an arbitrary arrival at time x such that $x \leq t$. Let q_x be the probability that this arriving entity is in the system at time t . Clearly,

$$q_x = 1 - G(t - x)$$

since it is the probability that this entity's service time is larger than $t - x$. Now consider a non-homogeneous Bernoulli splitting of the Poisson (arrival) process $\{N(t), t \geq 0\}$ such that with probability q_x an entity arriving at time x will be included in the split process. Since the split process counts the number in the $M/G/\infty$ system at time t , we have

$$p_j(t) = \exp \left\{ -\lambda \int_0^t q_x dx \right\} \frac{\left\{ \lambda \int_0^t q_x dx \right\}^j}{j!}.$$

The proof is described in Kulkarni [6], Gross and Harris [4] and Wolff [11]. It is based on conditioning the number of arrivals in time t and using the fact that each of the given arrivals occur uniformly in $[0, t]$. The argument is similar to the derivation of the number of departures from the $M/G/\infty$ queue in time t in Section 3.

Next we write down $p_j(t)$ in terms of entities given in the model, namely, λ and $G(t)$. As a result of using $q_x = 1 - G(t - x)$ and change of variables, we get

$$\begin{aligned} p_j(t) &= \exp \left\{ -\lambda \int_0^t q_x dx \right\} \frac{\left\{ \lambda \int_0^t q_x dx \right\}^j}{j!} \\ &= \exp \left\{ -\lambda \int_0^t [1 - G(t - x)] dx \right\} \frac{\left\{ \lambda \int_0^t [1 - G(t - x)] dx \right\}^j}{j!} \\ &= \exp \left\{ -\lambda \int_0^t [1 - G(u)] du \right\} \frac{\left\{ \lambda \int_0^t [1 - G(u)] du \right\}^j}{j!}. \end{aligned}$$

Using $p_j(t)$ we can also obtain

$$E[X(t)] = \lambda \int_0^t [1 - G(u)] du$$

and

$$\text{Var}[X(t)] = \lambda \int_0^t [1 - G(u)] du$$

by observing that for a given t , $X(t)$ is a Poisson random variable with parameter $\lambda \int_0^t [1 - G(u)] du$.

For the steady state analysis we let $t \rightarrow \infty$ in the transient analysis to derive all results. Let $X(t)$ converge in distribution to X as $t \rightarrow \infty$ such that $p_j(t) \rightarrow p_j$ where $p_j = \lim_{t \rightarrow \infty} P\{X(t) = j\}$. Taking the limit $t \rightarrow \infty$ for $p_j(t)$ we find

$$p_j = e^{-\lambda\tau} \frac{(\lambda\tau)^j}{j!}$$

where τ is the average service time. Then, with an integration-by-parts step it can be shown that

$$\tau = \int_0^\infty [1 - G(u)] du.$$

In addition the *mean and variance of the number in the system in steady state* are both $\lambda\tau$ since X is a Poisson random variable with parameter $\lambda\tau$. Note that the mean and variance of the sojourn times correspond respectively to the mean and variance of the service times since there is no waiting for service to begin. Before concluding this section, it is worthwhile observing that the steady-state probability p_j is identical to those of the $M/M/\infty$ system, and thus p_j does not depend on the CDF $G(\cdot)$ but just the mean service time.

2 Busy Period Distribution

As we described earlier, $\{X(t), t \geq 0\}$ is a regenerative process with regeneration epochs corresponding to times when the number in the system goes from 1 to 0. Each regeneration time corresponds to one idle period followed by one *busy period* (time when there are one or more entities in the $M/G/\infty$ system). Let U be the regeneration time and $U = I + B$, where the idle time $I \sim \exp(\lambda)$ (i.e. time for next arrival in a Poisson process), and the busy period B has a CDF $H(\cdot)$. We need to determine $H(\cdot)$. For that we develop the following explanation based on Example 8.17 in Kulkarni [6].

Let $F(\cdot)$ be the CDF of U . Using a renewal argument by conditioning on $U = u$, we can write down a renewal-type equation for $p_0(t) = P\{X(t) = 0 | X(0) = 0\}$ as

$$p_0(t) = \int_0^t p_0(t-u) dF(u) + \int_t^\infty P\{I > t | U = u\} dF(u).$$

Since $U = I + B$, $P\{I > t | U = u\} = 0$ if $u \leq t$, we can rewrite

$$\begin{aligned} p_0(t) &= \int_0^t p_0(t-u) dF(u) + \int_0^\infty P\{I > t | U = u\} dF(u) \\ &= \int_0^t p_0(t-u) dF(u) + P\{I > t\} \\ &= p_0 * F(t) + e^{-\lambda t} \end{aligned}$$

with $p_0 * F(t)$ being the standard convolution notation.

We already have an expression for $p_0(t)$ in Section 1 as

$$p_0(t) = \exp \left\{ -\lambda \int_0^t [1 - G(u)] du \right\},$$

where $G(\cdot)$ is the CDF of the service times. One way to obtain the unknown $F(t)$ function is to numerically solve for $F(t)$ in $p_0(t) = p_0 * F(t) + e^{-\lambda t}$ and similarly solve another convolution equation to get $H(t)$. An alternate approach, typically standard when there are convolutions, is to use transforms. We consider the Laplace Stieltjes Transform (LST) which is denoted as $\tilde{R}(s)$ for any positive-valued non-decreasing function $R(t)$ and is defined as

$$\tilde{R}(s) = \int_0^\infty e^{-st} dR(t).$$

Therefore, taking the LST on both sides of the equation $p_0(t) = p_0 * F(t) + e^{-\lambda t}$, we get

$$\tilde{p}_0(s) = \tilde{p}_0(s)\tilde{F}(s) + \frac{s}{s + \lambda}.$$

However, since we are ultimately interested in $H(t)$ we use the relation

$$\tilde{F}(s) = \left(\frac{\lambda}{\lambda + s} \right) \tilde{H}(s)$$

to obtain

$$\tilde{H}(s) = 1 + \frac{s}{\lambda} - \frac{s}{\lambda \tilde{p}_0(s)}.$$

In the most general case, the above LST is not easy to invert and obtain $H(t)$. Another challenge is to obtain $\tilde{p}_0(s)$ from $p_0(t)$ which is not trivial. However, there are several software packages available (such as Matlab and Mathematica) that can be used to numerically compute as well as invert LSTs.

Having said that, it is relatively straightforward to obtain $E[B]$, the mean busy period. Using the fact the $\tilde{p}_0(0) = p_0(\infty) = e^{-\lambda\tau}$, we can obtain

$$E[B] = -\tilde{H}'(0) = \frac{e^{\lambda\tau} - 1}{\lambda}.$$

Another way to obtain it is to use regenerative process results and solve for $E[B]$ in $1 - p_0 = \frac{E[B]}{E[B] + 1/\lambda}$.

3 Departure Process: Transient and Steady-State

Since the output from an $M/G/\infty$ may flow into some other queue, it is critical to analyze the departure process. Let $D(t)$ be the number of departures from the $M/G/\infty$ system in time $[0, t]$ given that $X(0) = 0$. For any arbitrary t , we seek to obtain the distribution of the random variable $D(t)$ and thereby characterize the stochastic process $\{D(t), t \geq 0\}$.

Similar to the analysis for the number in the system, here too we first consider transient and then describe steady-state results. The results follow the analysis in Gross and Harris [4]. However, it is crucial to point out that there are many other elegant ways of analyzing departures from $M/G/\infty$ queues and extending them, some of which we will see toward the end of this section.

If we are given that n arrivals occurred in time $[0, t]$, then using standard Poisson process results we know that the time of arrival of any of the n arrivals is uniformly distributed over $[0, t]$ and it is independent of the time of other arrival times. Therefore consider one of the n arrivals that occurred in time $[0, t]$. The probability $\theta(t)$ that this entity would have departed before time t can be obtained by conditioning on the time of arrival and unconditioning as

$$\theta(t) = \frac{1}{t} \int_0^t G(t-x)dx = \frac{1}{t} \int_0^t G(u)du.$$

In addition, the probability that out of the n arrivals in time $[0, t]$, exactly i of those departed before time t is $\binom{n}{i} [\theta(t)]^i [1 - \theta(t)]^{n-i}$.

Now, to compute the distribution of $D(t)$, we condition on $N(t)$ which is the number of arrivals in time $[0, t]$. In order to remind us that we do make the assumption that $X(0) = 0$ we include this condition in the expressions. Therefore we have

$$\begin{aligned} P\{D(t) = i | X(0) = 0\} &= \sum_{n=i}^{\infty} P\{D(t) = i | N(t) = n, X(0) = 0\} e^{-\lambda t} \frac{(\lambda t)^n}{n!} \\ &= \sum_{n=i}^{\infty} \binom{n}{i} [\theta(t)]^i [1 - \theta(t)]^{n-i} e^{-\lambda t} \frac{(\lambda t)^n}{n!} \\ &= \frac{[\theta(t)]^i}{i!} e^{-\lambda t} (\lambda t)^i \sum_{n=i}^{\infty} [1 - \theta(t)]^{n-i} \frac{(\lambda t)^{n-i}}{(n-i)!} \\ &= \frac{[\lambda t \theta(t)]^i}{i!} e^{-\lambda t \theta(t)}. \end{aligned}$$

Therefore for a given t , $D(t)$ is a Poisson random variable with parameter $\lambda t \theta(t)$. In other words, the transient distribution of the number of departure in time t is Poisson with parameter $\lambda t \theta(t)$. Having described the transient analysis, we just let $t \rightarrow \infty$ for steady-state analysis.

It can be shown from the definition of $\theta(t)$ that for very large t , $t\theta(t)$ approaches $t - \tau$. However, since we assume that τ is finite, as we let $t \rightarrow \infty$, the departure becomes a Poisson process with parameter $\lambda(1 - \tau/t) \rightarrow \lambda$. Of course we have not shown stationary and independence properties of the departure process, the reader is encouraged to refer to Serfozo [10] Section 9.6 for a derivation based on stationary marked point processes. Using the above results as well as stationary and independence properties, we can conclude that the departure process from an $M/G/\infty$ queue in steady state is the same as the arrival process, i.e. a Poisson process with parameter λ . Further, similar to the steady-state probability p_j being identical to that of the $M/M/\infty$ system and being independent of the CDF $G(\cdot)$, the departure process in steady state is also identical to that of the $M/M/\infty$ system and depends only on the mean service time (τ) and not its distribution.

We conclude this section by emphasizing that there is substantial amount of research on departure process from an $M/G/\infty$ queue that has not been covered here. Daley [2] shows that the departure process when viewed as a random translation of the arrival process is Poisson. That argument even works in the non-homogeneous case (i.e. the $M_t/G/\infty$ described in the next section). Also, a natural extension of output processes is to consider a network of $M/G/\infty$ queues. Analyzing a network of feedforward $M/G/\infty$ queues is particularly straightforward leveraging upon the fact that the output of an $M/G/\infty$ queue is also Poisson. For non-feedforward networks (i.e. networks with cycles) some of the properties are more tricky to show and the reader is referred to Serfozo [10]. Furthermore, it can be shown that (see Kobayashi and Mark [5]) the $M/G/\infty$ queue is quasi-reversible and hence useful in analyzing networks (for example to obtain product-form solutions).

4 Time-varying Arrivals: $M_t/G/\infty$

In this section we consider an extension to the $M/G/\infty$ queue. The arrival process is Poisson, however the parameter of the Poisson process is time varying. The average arrival rate at time t (for all t in $(-\infty, \infty)$) is a deterministic function of t represented as $\lambda(t)$. Hence the arrival process is defined as a non-homogeneous Poisson process. Everything else is the same as the regular $M/G/\infty$ queue. We call such a system an $M_t/G/\infty$ queue. This summary of results for the $M_t/G/\infty$ queue is based on Eick, Massey and Whitt [3].

Recall that we need to make one of the three assumptions stated in Section 1, otherwise we would need to know when service started for each of the customers at a reference time, say $t = 0$. We make the second assumption, although making one of the other two would not significantly alter the analysis but the results would not be as neat. In other words, assume that the system was empty at $t = -\infty$. Assume that for all finite t , $\lambda(t)$ is non-negative, measurable and integrable. With these assumptions in mind we state the results.

The results are in terms of the equilibrium random variable of the service times. In particular, let S be a random variable corresponding to a service time. We have already defined the CDF $G(x) = P\{S \leq x\}$ and mean $\tau = E[S]$. The equilibrium random variable S_e corresponding to the service times has a CDF $G_e(x)$ defined as

$$G_e(x) = P\{S_e \leq x\} = \frac{1}{\tau} \int_0^x [1 - G(u)] du.$$

One encounters equilibrium random variables in renewal theory. If a renewal process with inter-renewal random variable Y starts at $t = -\infty$, then at time $t = 0$, the remaining time for a renewal is according to Y_e .

Recall the notation $X(t)$. Here it denotes the number (of entities) in the $M_t/G/\infty$ system at time t . Then for any t , $X(t)$ has a Poisson distribution with parameter $\mu(t)$ given by

$$\mu(t) = E[\lambda(t - S_e)]E[S]$$

with $E[X(t)] = Var[X(t)] = \mu(t)$. It is important to note that $\lambda(\cdot)$ is the function defined in this section and not make the mistake of thinking that at $t = 0$ the average number in the system is negative! In fact, observe that the number in the system at any time depends

on the arrival rate S_e time units ago. Further, the departure process also has a similar time lag effect where the average departure rate at time t is $E[\lambda(t - S)]$ and the resulting process is non-homogeneous Poisson.

For $u > 0$ and any t ,

$$\text{Cov}[X(t), X(t + u)] = E[\lambda(t - (S - u)_e^+)]E[(S - u)^+]$$

where the notation $(y)^+$ is $\max(y, 0)$. In fact, this result can be derive for the homogeneous case (which we have not done earlier but is extremely useful especially in computer-communication traffic with long-range dependence). For an $M/G/\infty$ queue where $\lambda(t) = \lambda$,

$$\text{Cov}[X(t), X(t + u)] = \lambda P\{S_e > u\}E[S].$$

There are several results for networks of $M_t/G/\infty$ queues. Since the entities do not interact and departure processes are Poisson, the analysis is fairly convenient. The reader is referred to Eick, Massey and Whitt [3] as well as the references therein for further results.

References

- [1] C.S. Chang. *Performance Guarantees in Communication Networks*. Springer-Verlag, London, 2000.
- [2] D.J. Daley. Queueing output processes. *Adv. Appl. Probab.*, 8, 395-415, 1976.
- [3] S.G. Eick, W.A. Massey, and W. Whitt. The Physics of the $M_t/G/\infty$ Queue. *Operations Research*, Vol. 41, No. 4, 1993.
- [4] D. Gross and C.M. Harris. *Fundamentals of Queueing Theory*. 3rd Ed., John Wiley and Sons Inc., New York, 1998.
- [5] H. Kobayashi and B.L. Mark. *System Modeling and Analysis: Foundations of System Performance Evaluation*. Pearson/Prentice Hall, Upper Saddle River, NJ, 2008.
- [6] V.G. Kulkarni. *Modeling and Analysis of Stochastic Systems*. Texts in Statistical Science Series. Chapman and Hall, Ltd., London, 1995.
- [7] N. Likhanov and R.R. Mazumdar. Cell Loss Asymptotics in Buffers fed by Heterogeneous Long-tailed Sources. *In IEEE INFOCOM*, pp. 173-180, 2000.
- [8] M. Parulekar and A.M. Makowski. Tail probabilities for a multiplexer driven by $M/G/\infty$ input processes (I): Preliminary Asymptotics. *Queueing Systems - Theory & Applications*, Vol. 27, pp. 271-296, 1997.
- [9] S. Resnick and G. Samorodnitsky. Steady state distribution of the buffer content for $M/G/\infty$ input fluid queues. *Bernoulli*, Vol. 7, pp. 191-210, 2001.
- [10] R. Serfozo. *Introduction to Stochastic Networks*. Springer-Verlag, New York, NY, 1999.
- [11] R. W. Wolff. *Stochastic Modeling and the Theory of Queues*. Prentice Hall, Englewood Cliffs, N.J., 1989.